

Data Dictionary

Taxonomy (the classification hierarchy)

- **Category** -- Top-level product grouping (e.g., "Electrical", "Plumbing").

Column	Purpose
Category_ID	PK
Name	Category label
Description	Optional longer description
Active_Flag	1 = in use

- **ProductType** -- Sub-category under a Category (e.g., "Circuit Breakers" under "Electrical").

Column	Purpose
ProductType_ID	PK
Category_ID	FK → Category
Name	Product type label
Description	Optional longer description
UNSPSC_Code	Industry-standard classification code
Active_Flag	1 = in use

CategoryProductType -- Additional many-to-many link between Categories and ProductTypes (a ProductType can appear under multiple Categories).

Column	Purpose
Category_ID	FK → Category
ProductType_ID	FK → ProductType

Products

- **Product** -- Core product record. One row per part number.

Column	Purpose
Product_ID	PK
ProductType_ID	FK → ProductType (the assigned sub-category)
Manufacturer_ID	FK → Manufacturer
ProductName_ID	FK → ProductName (product line/series)
Product_Number	Part number
Product_Number_Custom	Customer-facing part number
Product_Name	Short name
Product_Description	Short description (up to 1750 chars)
Product_Active	1 = active
Deleted_Flag	1 = soft-deleted
Gross_Weight, Case_QTY, Unit_Quantity	Physical specs
<i>Price columns omitted</i>	

- **ProductDescription** -- Extended rich-text description for a product.

Column	Purpose
Product_ID	FK → Product (1:1)
Product_Description	Long description (up to 2000 chars)

- **ProductData** -- Additional product metadata and compliance flags.

Column	Purpose
Product_ID	FK → Product (1:1)
Product_Description_Pre	Pre-description text (1500 chars)
Product_Description_Post	Post-description text (250 chars)
TariffCode	Customs tariff code
Hazardous_Flag, Radioactive_Flag, Recycled_Flag, EnergyStar_Flag, GreenProduct_Flag, etc.	Compliance/classification flags

- **ProductDocuments** -- Linked spec sheets, images, PDFs per product.

Column	Purpose
Product_ID	FK → Product
DocumentTitle	Display name
DocumentLocation	URL/path to file

Column	Purpose
ImageFile	1 = image, 0 = document
DocumentOrder	Display order

Attributes

- **Attributes** -- Master list of attribute definitions (e.g., "Voltage", "Color", "Material").

Column	Purpose
Attribute_ID	PK
Name	Attribute label
Description	Optional description
FilterTypeID	FK → FilterType (how it's searched: range, exact, etc.)
Active_Flag	1 = in use

- **ProductTypeAttributes** -- Defines which attributes are expected for a given ProductType. This is the "schema" for each product type.

Column	Purpose
ProductType_ID	FK → ProductType
Attribute_ID	FK → Attributes
Display_Order	Ordering
IsPrimaryAttribute	1 = key identifying attribute
IsSmartAttribute	1 = used in smart search

Column	Purpose
Numerical	1 = numeric value expected
FilterFlag	1 = used as a search filter
DescFlag	1 = shown in description

- **ProductAttributeValues** -- The actual attribute values assigned to products. This is the labeled data.

Column	Purpose
Product_ID	FK → Product
ProductType_ID	FK → ProductType
Attribute_ID	FK → Attributes
Value	The attribute value (string, up to 60 chars)
DisplayText	Formatted display version
Suffix_ID	FK → Suffixes (unit of measure)
DigitalValue	Parsed numeric value (if applicable)
RangeLow, RangeHigh	Numeric range (if applicable)
Search_Text, KeyWords	Indexed search text
FilterFlag, DescFlag	Usage flags
IsShadowValue	1 = derived/computed, not directly entered

Reference/Lookup

- **Manufacturer** -- Brand/manufacturer.

Column	Purpose
Manufacturer_ID	PK
Name	Manufacturer name
MarketingDescription	Longer description

- **ProductName** -- Product line/series within a manufacturer (e.g., "Pro Series").

Column	Purpose
ProductName_ID	PK
ProductType_ID	FK → ProductType
Name	Line/series name

- **Suffixes** -- Units of measure (e.g., "VAC", "inches", "PSI").

Column	Purpose
Suffix_ID	PK
Name	Unit abbreviation
Description	Full unit name

- **FilterType** -- How an attribute is searched/displayed.

Column	Purpose
FilterTypeID	PK
Name	e.g., "Range", "Exact", "Text"

- **UNSPSC_Codes** -- Industry standard product classification codes.

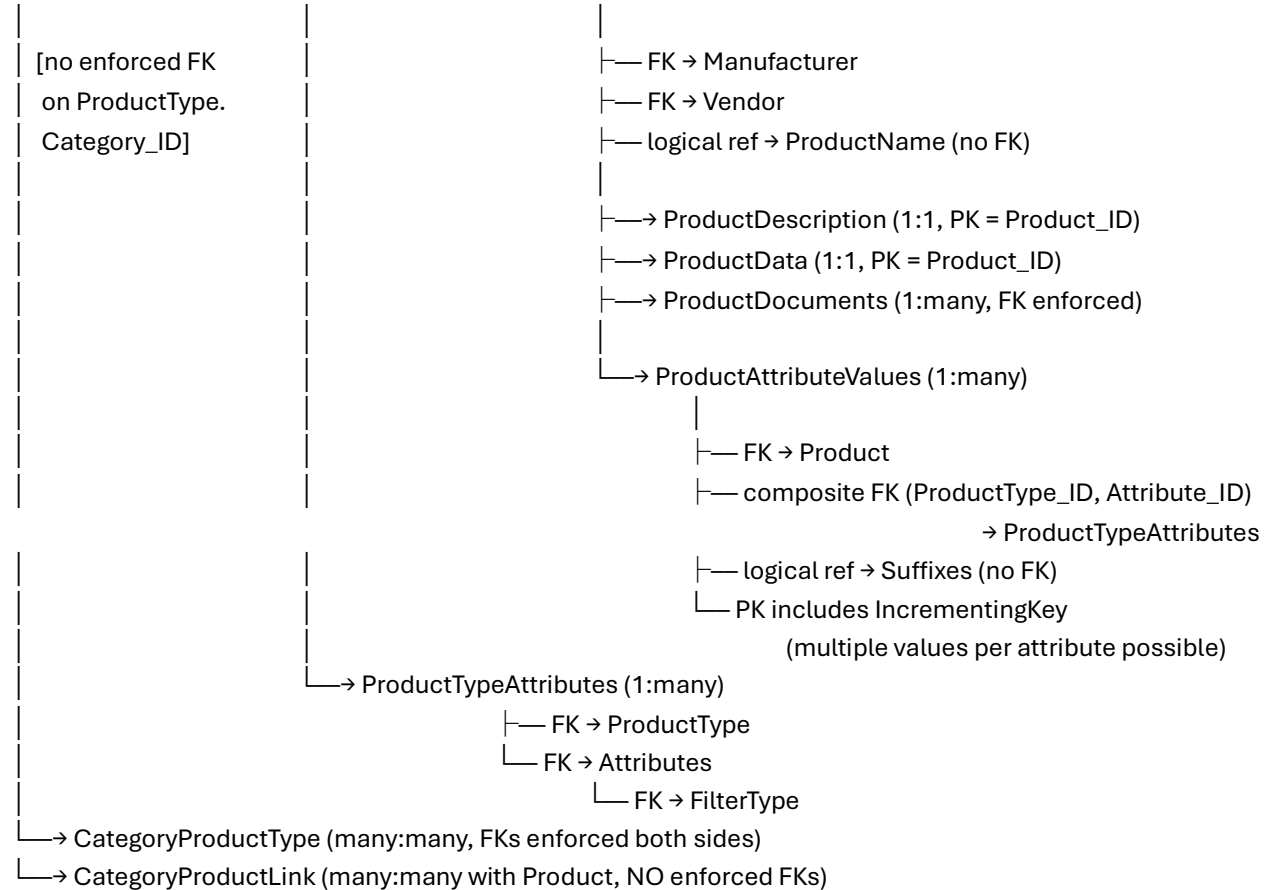
Column	Purpose
UNSPSC	The code
CategoryName, Hierarchy_Level_1, Hierarchy_Level_2	Classification hierarchy

- **CrossReference / CrossReferenceProduct** -- Alternate part numbers from other companies.

Column	Purpose
CrossReference_ID	PK
Company, ProductNumber	The external reference
Product_ID	FK → Product (via CrossReferenceProduct)

Relationship diagram (text)

Category (1) → (many) ProductType (1) → (many) Product



- ProductAttributeValues is the core labeled dataset. Its PK includes an IncrementingKey, meaning a single product can have **multiple values for the same attribute** -- don't assume one-to-one.
- Three joins are logical but not FK-enforced: ProductType.Category_ID, Product.ProductName_ID, and ProductAttributeValues.Suffix_ID. You may hit orphaned rows on those joins.

Alps stats

Metric	Value
Active Products	198465
Active Categories	77
Active Product Types	755
Active Attributes	487
Active Manufacturers	553
Products With Attribute Values	310608
Total Attribute Value Rows	4080763
Products With Descriptions	39979

Products With Documents	397261
Suffixes (Units of Measure)	53
UNSPSC Codes	8666
Cross References	89
ProductType-Attribute Mappings	1857
Synonyms	0
Category-Product Direct Links	3502
Avg Attributes Per Product	13.1
Avg Products Per ProductType	520.9
Avg ProductTypes Per Category	8.3
Avg Attributes Per ProductType	4.2